

1. INTEGRATED SITUATION

Scenario

SmartPark is company located in Rubavu District, western province of Rwanda. It provides multiple car-related services. It is struggling with inefficiencies due to its manual, paper based system for managing parking space sales. The human resource records employee details characterized by employee number, First name, Last name, address, position, Telephone, Gender, hired date and his or her department details characterized by code and department name. Based on department data human resource manually make payroll characterized by gross salary, total deduction, net salary and month of payment. This process is slow, prone to errors and makes it difficult to tack employee's payment and generate monthly report efficiently. To address these challenges, the SmartPark needs a web based application that handles the Payroll making process. The system should allow the human resource to record employee details digitally and automatically generate needed reports.

Task: Employee Payroll Management System (EPMS)

As a full stack developer, you are given 7 hours to develop that web based application by:

1. Using attributes provided below, design an Entity Relationship Diagram (ERD) that represents the relationship between their entities.
 - Identify appropriate **primary keys** and **foreign keys** based on the relationships among the entities.
 - Entities and attributes are:
 1. Employee (employeeNumber, FirstName, LastName, Position, Address, Position, Telephone, Gender, hiredDate)
 2. Department (DepartementCode, DepartementName, GrossSalary)
 3. Salary (GlossSalary, TotalDeduction, NetSalary, month)
 - ERD should be drawn before using the computer.
 - ERD should be drawn on plain paper using pencils.
 - ERD should indicate cardinalities, relationships with correct symbols

2. Creating database called EPMS with Employee, Department and Salary tables as designed in ERD.

➤ Use this information for departments and Payroll marking

DepartementCode	DepartementCode	GrossSalary	Total deduction
CW	Carwash	300,000 rwf	20,000 rwf
ST	Stock	200,000 rwf	5,000 rwf
MC	Mechanic	450,000 rwf	40,000 rwf
ADMS	Administation Staff	600,000 rwf	70,000 rwf

3. Saving your work in your real names in a folder called (FirstName_LastName_National_Practical_Exam_2025).
4. Preparing React.js Front-end application development environment by installing required modules and dependencies
5. Preparing JavaScript runtime environment for Node.js
6. Creating react.js components with UI features that will enable user to input his data in the tables above and display required reports.
 - Menu bar/Pages of web application should include Employee, Department and Salary, Reports and Logout options.
 - The web application should be responsive
7. Creating react.js components with UI features that will enable user to input his data in the tables above and display required reports.
8. Developing backend and frontend of parking sales management system.
 - Name backend project folder as **backend-project** and frontend project folder as **frontend-project**.
 - Use Tailwind CSS to implement UI design.
 - Your backend application should communicate to MySQL database/MongoDB to perform CRUD operations of using Node.js runtime environment and express.js framework.

Note: 1. Insert operation should be used on all four (4) forms (Employee, Department and Salary)

2. Delete, update, retrieve operations should only be used on
Salary form

9. Creating react.js components with UI features that will enable user to input his data in the tables above and display required reports.
10. Creating a session based login user account having username and password.
11. Integrating your react.js application to backend application using Axios to input his data in the tables above and display required reports.
12. Generating monthly Employ payroll composed FirstName, LastName, Position, Departement and Net Salary
13. Removing permanently your project with its related configurations after being marked.
 - Ask permission to the assessor before removing your project.

Note to the assessors:

- An assessor may promptly intervene to prevent the unnecessary use of internet and any other injury/incident that may occur during assessment
- The assessor should allow lunch time (one hour) break for candidate
- The assessor should ensure that each candidate is given time to exhibit his/her project by answering the following question:
 1. Describe your product in brief.
 2. Which difficulties have you met along the implementation of the project
 3. How did you overcome the difficulties
- The assessor should make sure that the distance between candidates prevents cheating
- The assessor should collect all ERD drawn by the candidates together with the exported project as evidences.